

Element J

Our stakeholder is Mr. Kiser who has a bachelor of science degree in chemical engineering. His main subjects are physical science, chemistry, physics, and engineering. Mr. Kiser is the main stakeholder and client because he is the one who discovered this problem. He has been experiencing back pains while trying to push the cart from a handle that is just too low for him. He is our engineering service learning teacher who has been testing our designs as we have been making iterations during the past couple of weeks. One day, we interviewed him with his problems about the cart and he responded to our questions. One of our teammates recorded this interview. We used his responses to create the criteria and restraints for this design such as adjustability, longevity, budget friendly, etc. For instance, near the start of the interview, Mr. Kiser pointed out that when he pushes the cart he ends up "arching the back . . . the lower back just naturally curves and I think that's where a lot of stress/strain occurs . . ." This points out the severity of the problem for our stakeholder and the reason that we are trying to solve the problem.

Link to voice memo of interview with answers to questions by our class:

 Kiser criteria.m4a

Mrs. Klein-Gardner leads the engineering for USA (E4USA) program that is the basis of the class we are in as we create this project. She comes a couple of Mondays throughout the year and she even introduced some experts in engineering from Vanderbilt. Some of these experts that helped us along the way were Dr. London. She is an associate professor

of mechanical engineering at Vanderbilt University. She was the one who helped us consider a stronger and more stable base because she is in mechanical engineering. Also she gave the helpful idea to add small instructions for disassembly of the product. The reason she is able to have this mindset is because she has a background in human factors engineering. Also, Jared, a student at Vanderbilt, is in the medical engineering field. With this engineering background, he was able to give us advice for a complex lock mechanism that we will be sure to include as more iterations are made in the future.

The combination of the criteria from Mr. Kiser helped us have a better idea of what problem we are trying to solve. Together, with the help of the specialists from Vanderbilt, we were able to create our newest prototype in time to present to our stakeholder. Our project will be able to help our client hopefully reduce back pain and make the overall experience of pushing his cart easier.